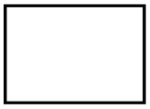
NPTEL Online Certification Course
Industrial Engineering and Operations Research
Assignment 2: Detailed Solution
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Q.1. Match the symbol with its correct description

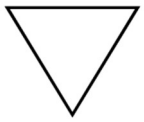
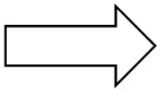
Symbol		Operation	
P		1	Putting cap
Q		2	Storage
R		3	Inspection
S		4	Transportation

(A) P-3, Q-2, R-1, S-4 (B) P-1, Q-3, R-3, S-2 (C) P-3, Q-1, R-4, S-2 (D) P-2, Q-1, R-3, S-4

Sol. Option (C) is correct.

Symbol	Meaning	Description
	Putting cap (operation)	Represents an operation in the process.
	Inspection	Denotes an inspection for either quantity or quality in the process. The quantity inspection may consist of measuring, counting, or weighing. A quality inspection may consist of testing the part to see whether it meets a predetermined standard.

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	Storage	Represents storage where materials or objects are kept before the next operation. Example: warehouse storage.
	Transportation	Indicates movement of material or object from one place to another. Example: Conveyor belt transport.

Q.2. Which of the following statements is/are TRUE?

(A) Work study comprises method study and time study.
(B) Method study is a part of work measurement but time study is a part of motion study.
(C) Motion study comprises method study and time study.
(D) Method study is a motion study and time study is a part of work measurement.

Sol. Work study comprises two areas of study– method study (motion study) and time study (work measurement). Method study focuses on analyzing and improving work methods — it involves motion study, which looks at the physical movements in a task to eliminate unnecessary ones. Time study is a key part of work measurement, which aims to determine the time required to perform a job efficiently. Hence, options (A) and (D) are correct.

Q.3. Out of 2000 observations, a worker was found working on 750 occasions. Some z-values are given in the following table:

Confidence level	90%	95%	99%	99.9%
z value	1.645	1.960	2.576	3.291

For 90% confidence level, the upper bound of the proportion of the time worker works is
(A) 0.375 (B) 0.357 (C) 0.392 (D) 0.385

Sol.

Given,

Number of observations, n =2000

Number of times worker was working = 750

Hence, sample proportion $\bar{p} = \frac{750}{n} = \frac{750}{2000} = 0.375$

At 95% confidence level, z = 1.645 (given)

The standard error of a proportion is given by

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$$s_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = \sqrt{\frac{0.375(1-0.375)}{2000}} = 0.0108$$

Margin of error, $M_E = z \times s_p = 1.645 \times 0.0108 = 0.0177$

Therefore, upper bound = $\bar{p} + M_E = 0.375 + 0.0177 = 0.3927$

Hence, option (C) is correct.

Q.4. A teacher has set a question paper and asked five students to solve it. The time taken by the students and their corresponding CPIs are given in the table. Assign a performance rating in the following manner: 1.5 (CPI 9–10), 0.6 (CPI 5–6); the performance rating is a linear function of CPI in the CPI range 6–9. Providing 10% allowance on the normal time, find out the average standard time.

Student	CPI	Time taken (Minutes)
Vedant	9.3	120
Payal	8.1	130
Mathews	7.8	140
Ajay	5.6	180
Mohsin	6.4	180

(A) 147.42 (B) 150 (C) 162.16 (D) 118.18

Sol.

Performance Rating (R)

Given:

CPI 9–10 $\rightarrow R = 1.5$

CPI 5–6 $\rightarrow R = 0.6$

Between 6–9 $\rightarrow$ linear:

$$R = 0.6 + \frac{(CPI - 6)}{3} \times (1.5 - 0.6) = 0.6 + 0.3 \times (CPI - 6)$$

Now computing for each student:

Student	CPI	Rating (R)
Vedant	9.3	1.5
Payal	8.1	$0.6 + 0.3(2.1) = 1.23$
Mathews	7.8	$0.6 + 0.3(1.8) = 1.14$
Ajay	5.6	0.6
Mohsin	6.4	$0.6 + 0.3(0.4) = 0.72$

Normal Time

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$$Normal\ time = Observed\ time \times R$$

Student	Time	Rating	Normal Time
Vedant	120	1.5	180
Payal	130	1.23	159.9
Mathews	140	1.14	159.6
Ajay	180	0.6	108
Mohsin	180	0.72	129.6

Standard Time= Normal time+ 10% allowance on normal time

$$Standard\ time = Normal\ time \times 1.10$$

Student	Standard Time
Vedant	198.0
Payal	175.89
Mathews	175.56
Ajay	118.8
Mohsin	142.56

Average Standard Time

$$Average = \frac{198 + 175.89 + 175.56 + 118.8 + 142.56}{5} = 162.16\ Minute$$

Hence, option (C) is correct.

Q.5. Work sampling was introduced by

(A) F.W. Taylor (B) Henry Ford (C) Walter A. Shewhart (D) L.H.C. Tippet

Sol. Work sampling was introduced by L.H.C. Tippet, a British statistician. It estimates the proportion of time that a worker spends working. Hence, option (D) is correct.

Q.6. In a survey of 200 people, 60 people preferred a new product. What are the sample proportion $\bar{p}$ and its standard error s_p, respectively?

(A) 0.30, 0.032 (B) 3.33, 0.046 (C) 3.33, 0.032 (D) 0.30, 0.046

Sol.

We are given:

number of successes, $x = 60$

total observations, $\bar{p} = 200$

Step 1: Sample proportion

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$$\bar{p} = \frac{x}{n} = \frac{\text{number observed in classification}}{\text{Total number of observations}} = \frac{60}{200} = 0.30$$

Step 2: Standard error

$$s_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = \sqrt{\frac{0.30 \times 0.70}{200}} = 0.032$$

Hence, option (A) is correct.

Q.7. Which one of the following is a bad practice in computer use from ergonomics point of view?

(A) Positioning the monitor so that the top of the screen is at or slightly below eye level
(B)Maintaining wrists in a neutral position while typing
(C) Using a chair that supports the natural curve of the spine
(D) Keeping the keyboard elevated above elbow level to improve visibility

Sol. Correct practice: The monitor should be at or slightly below eye level to reduce neck strain. Wrists should remain in a neutral (straight) position to avoid strain. A chair that supports the spine helps maintain proper posture. Bad practice: The keyboard should be at or slightly below the elbow level, not above. Raising it can cause wrist and shoulder strain. Hence, option (D) is correct.

Q.8. Which light filter is often used in smartphones?

(A) Red light filter (B) Yellow light filter (C) Blue light filter (D) Violet light filter

Sol. A blue light filter is a feature or tool designed to block the amount of blue light (≈400–495 nm) emitted by screens such as those on smartphones, computers, and tablets. Hence, option (C) is correct.

Q.9. Choose all applicable options. Inventory is needed to

(A) provide a selection of goods for anticipated customer demand and to separate the firm from fluctuations in that demand
(B) hedge against inflation and upward price changes
(C) decouple various parts of the production process
(D) take advantage of quantity discounts, because purchases in larger quantities may reduce the cost of goods or their delivery.

Sol. The functions of Inventory are the following:

- To provide a selection of goods for anticipated customer demand and to separate the firm from fluctuations in that demand
- To hedge against inflation and upward price changes
- To decouple various parts of the production process
- To take advantage of quantity discounts, because purchases in larger quantities may reduce the cost of goods or their delivery

Hence, options (A), (B), (C) and (D) are correct.

Q.10. Which one of the following is not considered an inventory?

(A) Maintenance/repair/operating supply (MRO) (B) Work-in-progress (WIP)

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(C) Finished goods (D) Capital equipment

Sol. Maintenance/repair/operating supply (MRO), work-in-progress (WIP), and finished goods are common types of inventories held in production and supply chains. However, capital equipment refers to long-term assets like machinery or buildings, and is not considered an inventory. Hence, option (D) is correct.

Q.11. The annual demand of an item is 7000 units. The basic EOQ model provides economic order quantity as 450. If the demand increases to 13000 units, the economic order quantity is

(A) 836 (B) 936 (C) 900 (D) 614

Sol. Economic Order Quantity is given by

$$EOQ = \sqrt{\frac{2DS}{H}}$$

where S is ordering cost, and H is holding cost.

Calculate the new EOQ using proportionality:

$$\frac{EOQ_2}{EOQ_1} = \sqrt{\frac{D_2}{D_1}} = \sqrt{\frac{13000}{7000}} = \sqrt{1.857}$$

$$\Rightarrow EOQ_2 = EOQ_1 \times \sqrt{1.857} = 450 \times \sqrt{1.857} \approx 614$$

Hence, option (D) is correct.

Q.12. A company produces a product at a rate of 400 units per day. The daily demand for the product is 200 units. The setup cost per production run is ₹150, and the holding cost per unit per year is ₹25. Assuming 280 working days in a year, the optimal production order quantity is

(A) 1300 units (B) 600 units (C) 820 units (D) 1160 units

Sol.

Given,
Annual demand, $D = 200 \times 280 = 56{,}000$ units
Setup cost, $S = ₹150$
Holding cost per unit per year, $H = ₹25$
Daily production rate, $p = 400$ units/day
Daily demand rate, $d = 200$ units/day

Optimal Production Order Quantity is given by

$$EOQ = \sqrt{\frac{2DS}{H\left(1-\frac{d}{p}\right)}} = \sqrt{\frac{2 \times 56000 \times 150}{25 \times \left(1-\frac{200}{400}\right)}} = 1159.3 \approx 1160$$

Hence, option (D) is correct.

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Q.13. Which of the following is the key factor in the quantity discount model?

(A) Constant demand (B) Uncertain lead times
(C) Price breaks based on quantity (D) Random demand

Sol. The quantity discount model evaluates the impact of order quantity on total cost when prices are set at different levels, meaning lower unit prices are offered for larger order quantities. These are known as price breaks or quantity discounts. Hence, option C is correct.

Q.14. When using the quantity discount model, the optimal order quantity is determined by

(A) minimizing total holding cost
(B) balancing holding, ordering, and price discount costs
(C) maximizing price discount
(D) minimizing ordering cost

Sol. In the quantity discount model, the goal is to find the optimal order quantity that minimizes the total cost, which includes ordering costs (cost per order), holding costs (cost to store inventory), and price discount costs. Larger order quantities give better price discounts, but they also increase holding cost. However, smaller order quantities reduce holding costs, but lead to more frequent orders (ordering cost increases) and may miss out on discounts. Thus, the optimal quantity is found by balancing holding, ordering, and price discount costs. Hence, option (B) is correct.

Q.15. The lead time for getting an item is 12 days. Daily demand is 24 units. Safety stock is 40 units. The reorder point is

(A) 14 units (B) 972 units (C) 504 units (D) 328 units

Sol.

Given,
Lead time = 12 days
Daily demand = 24 units
Safety stock = 40 units

Reorder Point is given by

ROP = (Daily demand × Lead time) + Safety stock = (24×12) + 40 = 288 + 40 = 328 units.

Hence, option (D) is correct.

Q.16. A newspaper seller sells 500 copies of a newspaper each day. He pays Rs. 7 for each paper and sells for Rs. 10. He is given a Rs. 2 credit for each unsold paper. The respective costs of shortage and overage per unit are
(A) Rs. 3, Rs. 7 (B) Rs. 3, Rs. 2 (C) Rs. 3, Rs. 5 (D) Rs. 7, Rs. 5

Sol.

Given,
Cost price, CP = ₹7
Selling price, SP = ₹10
Credit amount for unsold paper = ₹2

Cost of shortage

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If a customer comes but no paper is available, the seller misses ₹10 sale. Since, the seller didn't buy the paper in the first place, he saves ₹7 (CP).

Thus, costs of shortage per unit = SP – CP = 10 – 7 = ₹3, i.e., the seller loses the ₹3 profit.

Cost of overage

If a paper is left unsold, the seller gets ₹2 credit back as refund. Since, the seller paid ₹7 to buy, cost of overage per unit = CP – credit amount = 7 – 2 = ₹5, i.e., the seller loses ₹5.

Hence, option (C) is correct.

Q.17. The quote "*Competition is becoming less between individual companies, and more between the supply chains they are part of*" is attributed to

(A) Richard Wilding (B) J.M. Juran (C) G. Taguchi (D) T. Ohno

Sol. The quote "*Competition is becoming less between individual companies, and more between the supply chains they are part of*" is attributed to Richard Wilding. He is a well-known professor of supply chain strategy (at Cranfield School of Management, UK) and a prominent voice in modern logistics and supply chain thinking. Wilding emphasised that businesses are no longer isolated entities; they operate as part of interconnected networks, and performance depends on how well the entire chain is coordinated. Hence, option (A) is correct.

Q.18. When was the term "Supply Chain Management" first used
(A) 1982 (B) 1978 (C) 1988 (D) 1992

Sol. An The term "Supply Chain Management" was first used in 1982 by Keith Oliver, a business consultant in an interview. In the 90s, this term gained popularity.

Hence, option (A) is correct.

Q.19. A supply chain phenomenon where small fluctuations in demand at the retail level cause progressively larger fluctuations at the wholesale, distributor, and manufacturer levels, leading to inefficiencies and increased costs is called
(A) Muda effect (B) Mura effect (C) Muri effect (D) Bullwhip effect

Sol. The Bullwhip effect refers to the supply chain issue or phenomenon where small fluctuations in consumer demand cause larger and amplified fluctuations in orders at each upstream level (retailer→wholesaler→distributor→manufacturer), resulting in inefficiencies, excess inventory, and higher costs. Hence, option (D) is correct.

Q.20. Which of the following statements are true in the aspects of comparison of MRP and MRP–II?
(A) MRP focuses on Material planning, whereas MRP–II focuses on Comprehensive resource planning.

(B) MRP confines to production needs, whereas MRP–II includes financial and human resource (HR) aspects.

(C) MRP can be a standalone system, whereas MRP–II is integrated with ERP systems

(D) MRP can have real-time feedback, whereas MRP–II has no Real-time adjustments.

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Sol. MRP focuses on Material planning, whereas MRP–II focuses on Comprehensive resource planning. MRP have scope on production needs, whereas MRP–II focuses have scope on financials, HR, etc. MRP can have Integration as a standalone system, whereas MRP–II is integrated with ERP systems. Conventional MRP has no real-time feedback, whereas MRP–II has Real-time adjustments possible.

Hence, options (A), (B) and (C) are correct.

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